CH 4
ACTUATORS



ACTUATORS

In this Chapter we will present:

- Direct current (DC) Motor
- H-Bridge (Motor Interface)
- Pulse-Width-Modulation (PWM)
- Servos
- Stepper Motors



ACTUATORS FOR ROBOTS

- Actuators are used in order to produce mechanical movement in robots.
- Its types are :
  - Electrical
  - Hydraulic
  - Pneumatic
  - Others



TYPES OF ACTUATORS

The most common actuators are:

- **Electric motors**
  - ✓ The most common actuators in mobile robots
  - ✓ It is an electric machine that converts electrical energy into mechanical energy.

- **Pneumatic and hydraulic actuators**,
  - ✓ Used in industry for large manipulation tasks but seldom for mobile robots
  - ✓ Use compressed air/oil to produce motion
  - ✓ Used for lower cost and lower precision applications



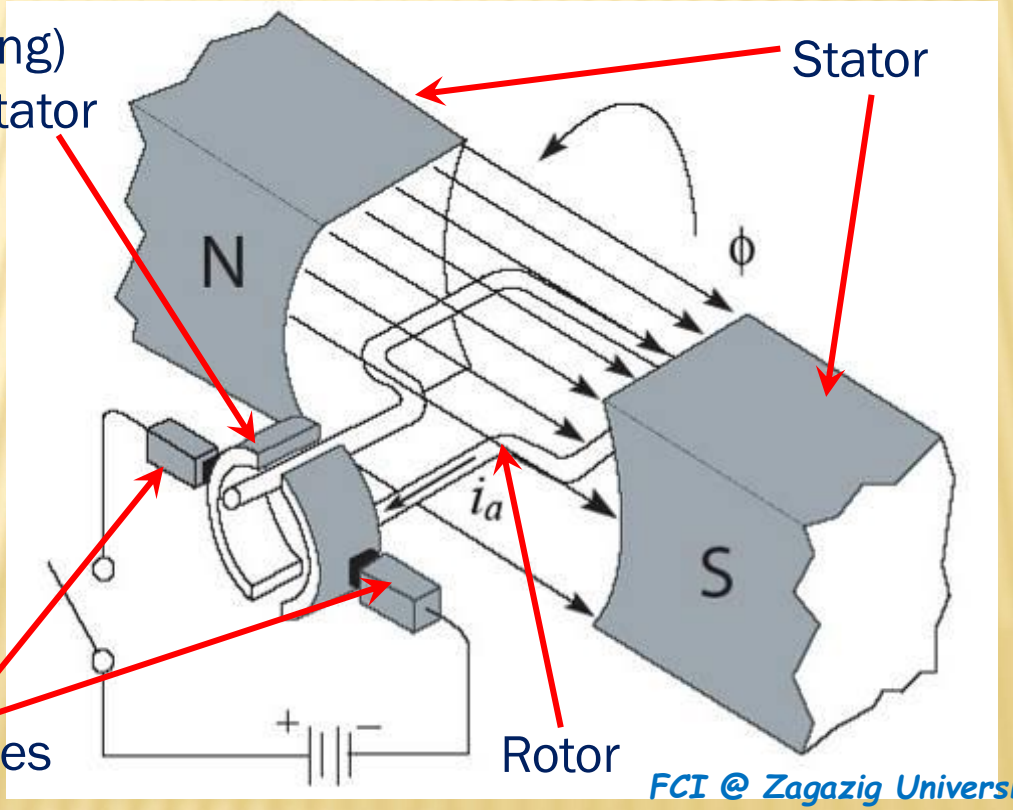
ELECTRICAL ACTUATOR TYPES

- DC-motors
- AC-motors
- Stepper motors
- Servos



ELECTRIC DC MOTORS

- Electric motors are the most common source of torque for mobility and/or manipulation in robotics
- The physical principle of all electric motors is that when an electric current is passed through a conductor (usually a coil of wire) placed within a magnetic field, a force is exerted on the wire causing it to move



COMPONENTS OF DC MOTOR

The principle components of an electric motor are:

- **North and south magnetic poles**
  - Provide a strong magnetic field.
  - Being made of bulky ferrous material they traditionally form the outer casing of the motor called **stator**
- **An armature (Rotor)**

  which is a cylindrical ferrous core rotating within the stator and carries a large number of windings made from one or more conductors

COMPONENTS OF DC MOTOR

- **Commutator**,

  which rotates with the armature and consists of copper contacts attached to the end of the windings

- **Brushes**
  - in fixed positions and in contact with the rotating commutator contacts.
  - They carry direct current to the coils, resulting in the required motion

HOW DC MOTOR WORKS

https://www.youtube.com/watch?v=j_F4IimaHYI

- The classic DC motor has a rotating armature in the form of an electromagnet
- A rotary switch called a commutator reverses the direction of the electric current twice every cycle, to flow through the armature so that the poles of the electromagnet push and pull against the permanent magnets on the outside of the motor
- As the poles of the armature electromagnet pass the poles of the permanent magnets, the commutator reverses the polarity of the armature electromagnet.
- During that instant of switching polarity, inertia keeps the motor going in the proper direction

GEARED DC-MOTORS

The speed of DC motor ranges from 1000 to 5000 revolution per minute (rpm).

This speed is very high to be used in mobile robots

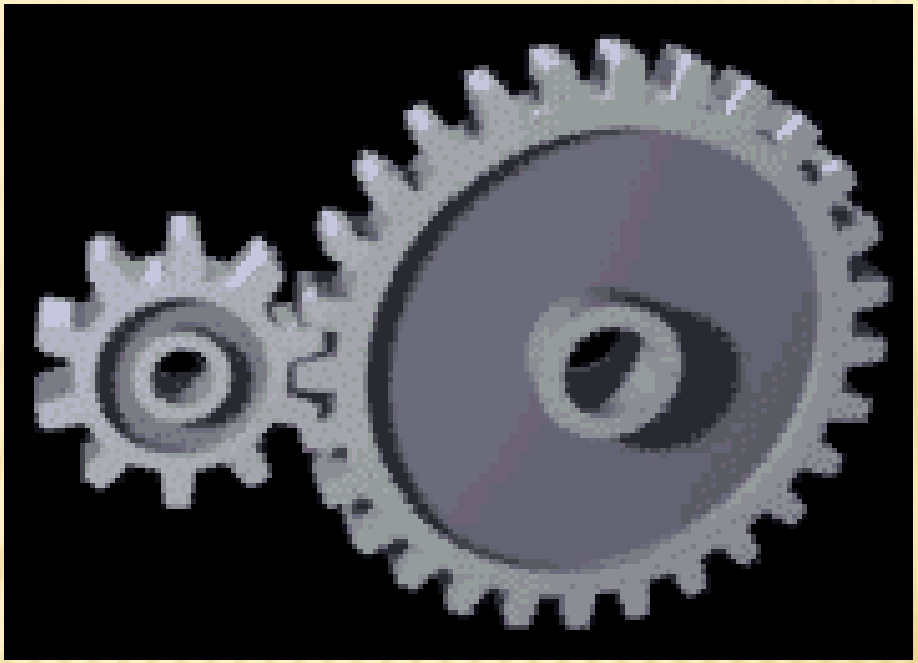
There are two ways to reduce the speed

- ✓ using gearbox
- ✓ reducing deriving power (voltage/current)

Gears are used to reduce/increase the speed depending on gear ratio



GEARED DC-MOTORS

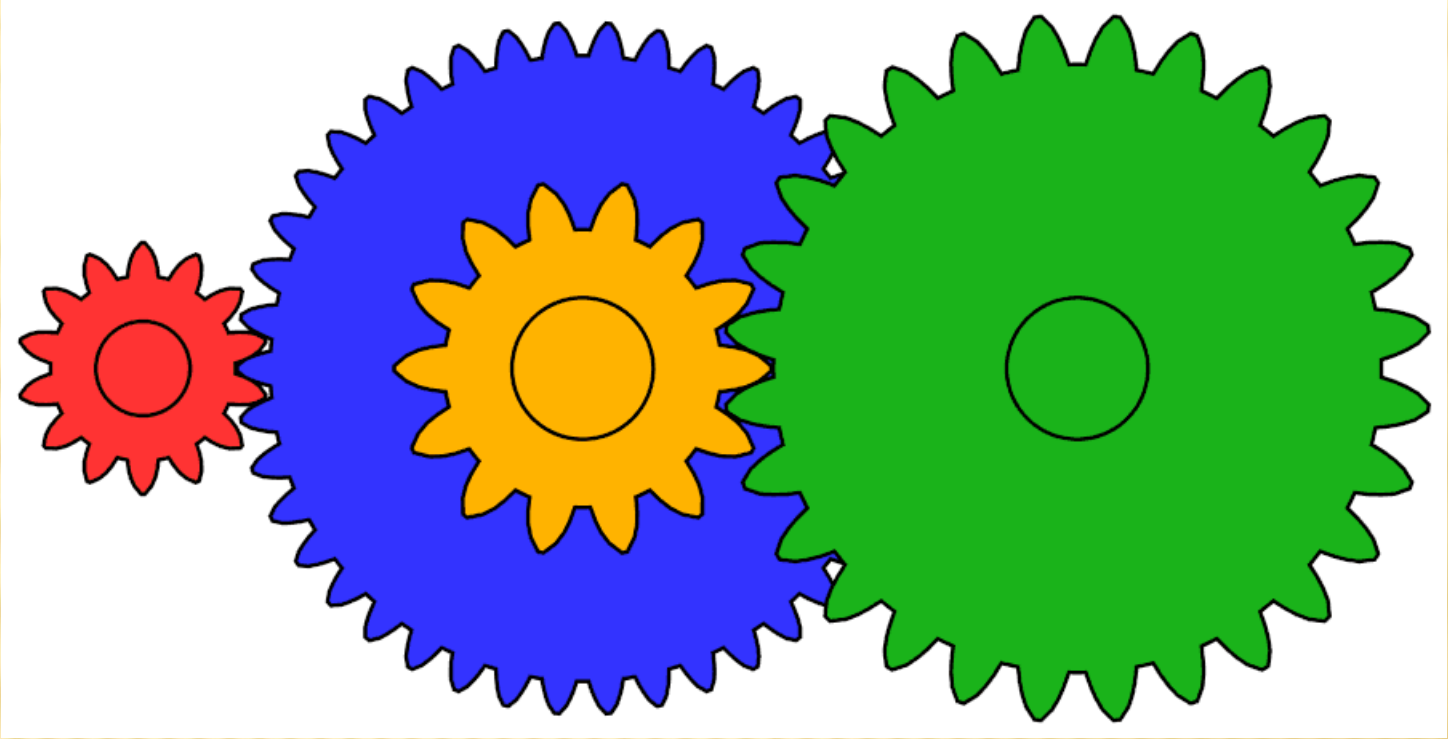


- Speed reduction is produced If the motor shaft is coupled to the smaller gear and the final movement is taken from the bigger gear
- Speed will increase if the motor shaft is coupled to the bigger gear
- Amount of speed reduction is proportional to N1/N2 where

  N1 is number of gear1's teeth

  N2 is the number of gear2's teeth

GEARED DC-MOTORS



- To produce high reduction ration a multiple gears arrangement is used as shown
- This arrangement is called GearBox
- The Torque/Force of the movement is increased as speed is reduced by gearbox

ENCAPSULATED DC-MOTORS





- Encapsulated motor is a single package that includes dc-motor, gearbox and even the shaft encoder

ADVANTAGES OF DC-MOTORS

- Simple
- Cheap
- Easy to control

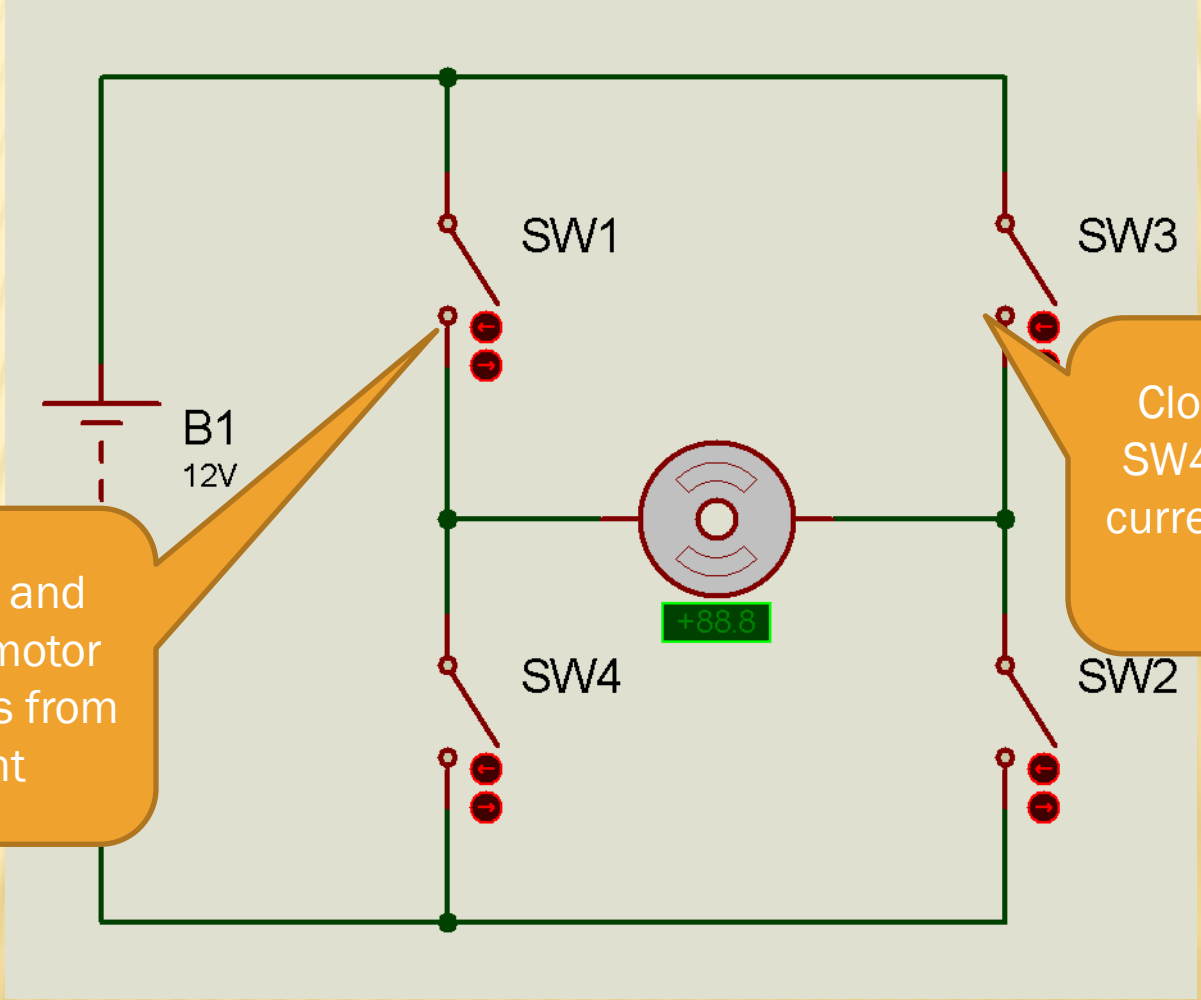


H-BRIDGE DEFINITION:

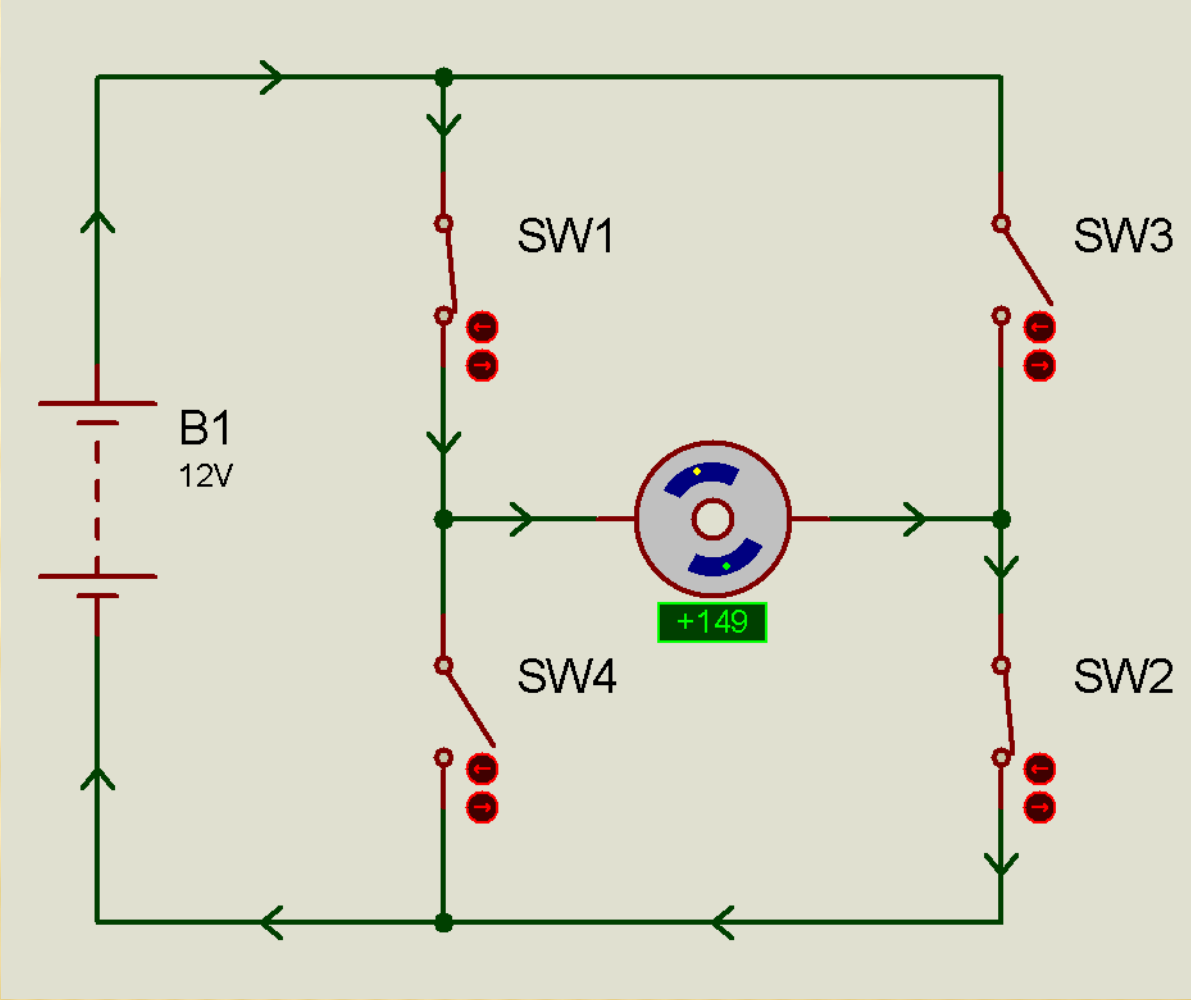
- An **H bridge** is an electronic circuit that allows a motor to be driven in both direction
  - ✓ needed to run a motor forward and backward
  - ✓ Clockwise or Counter Clockwise
- It Consists of 4-electronic switches in a bridge configuration



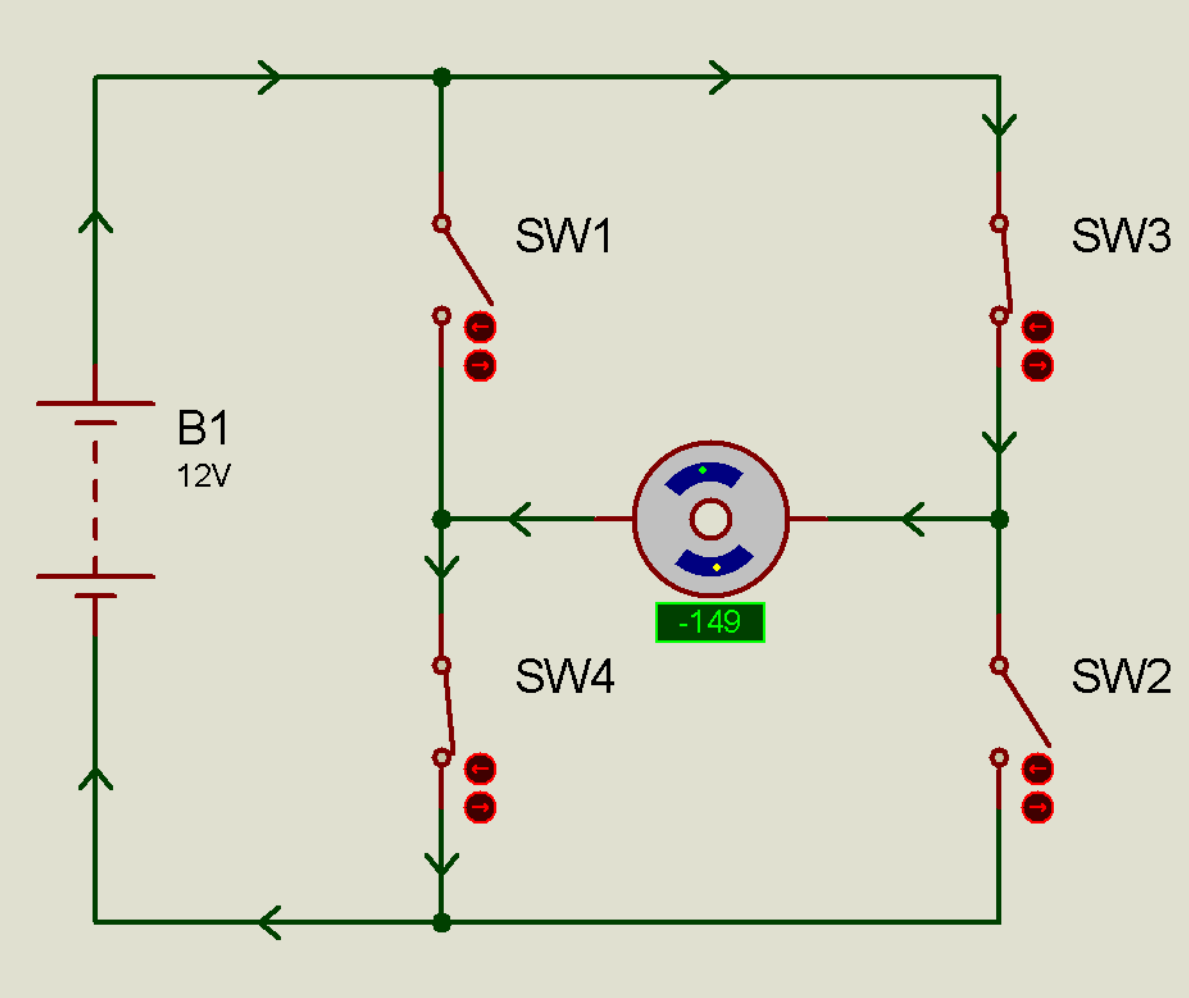
H-BRIDGE OPERATION



H-BRIDGE OPERATION FORWARD

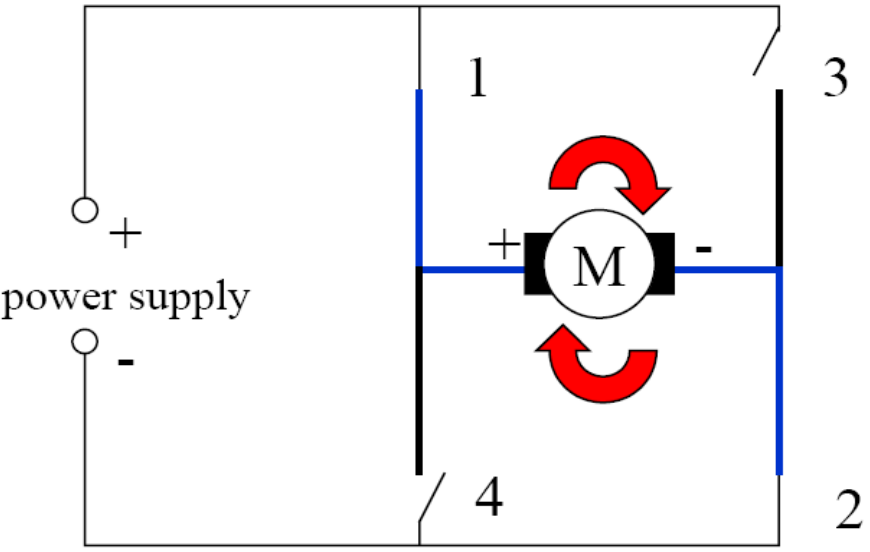


H-BRIDGE OPERATION REVERSE

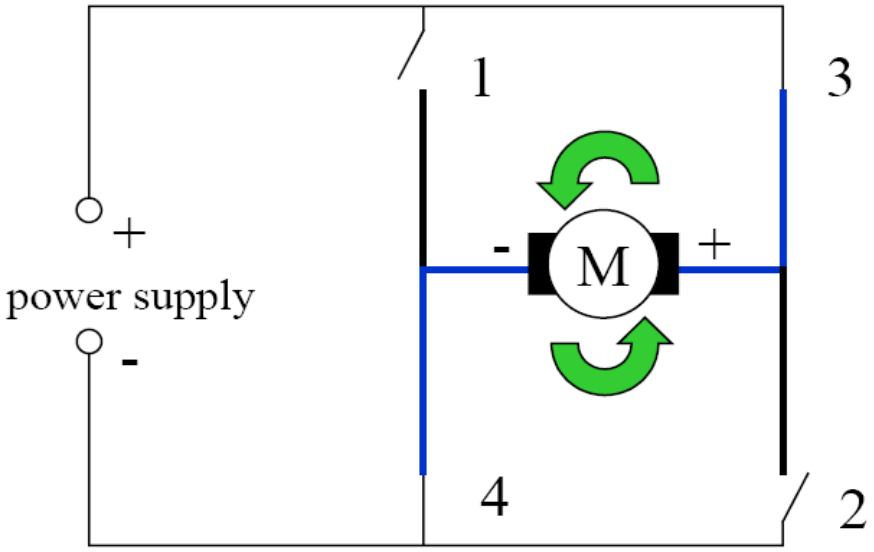


H-Bridge

Drive forward:

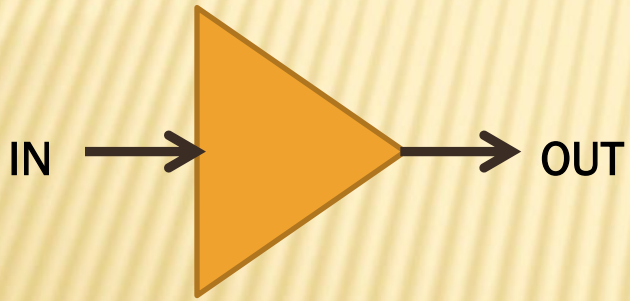


Drive backward:

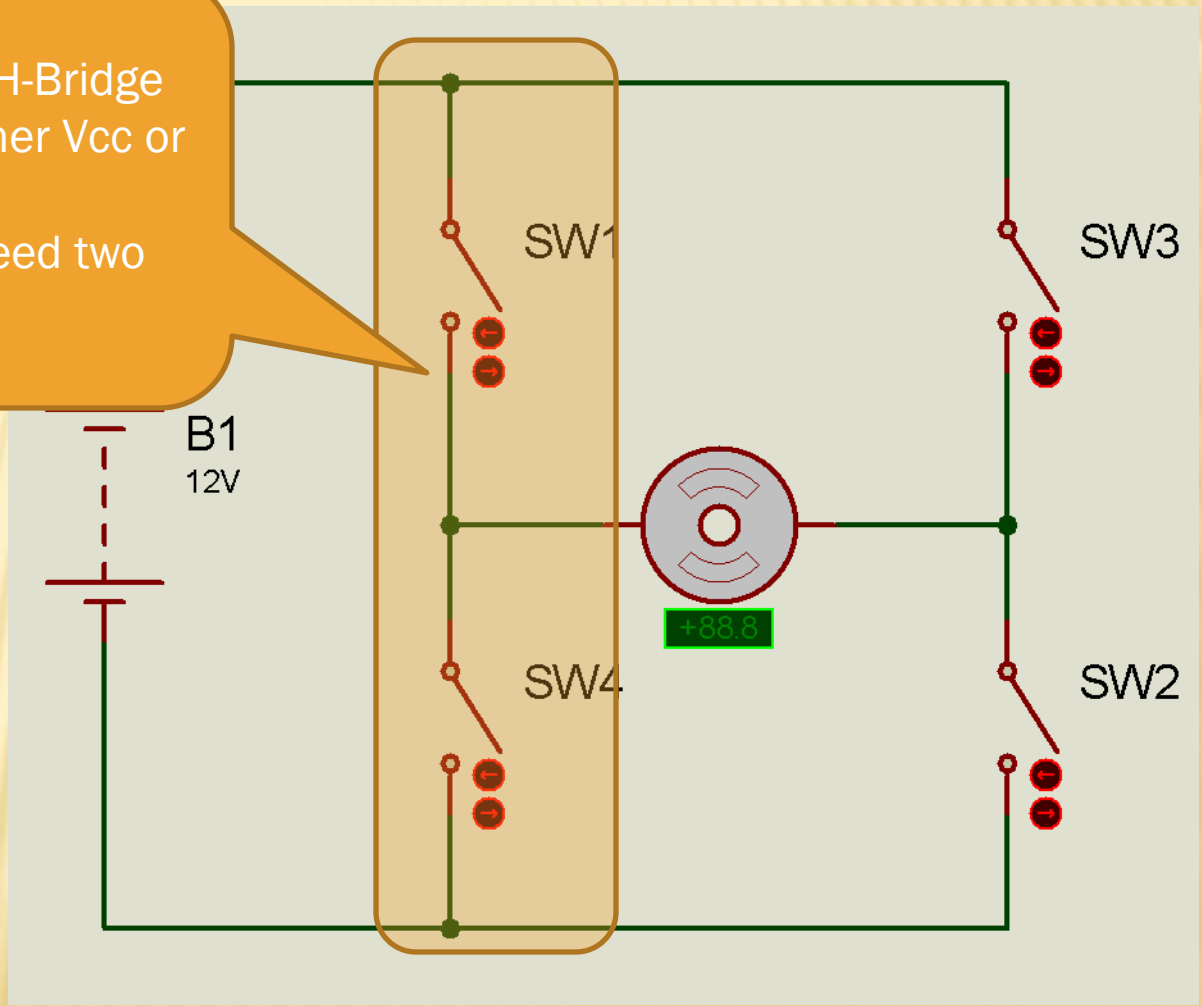


FULL VS HALF H-BRIDGE

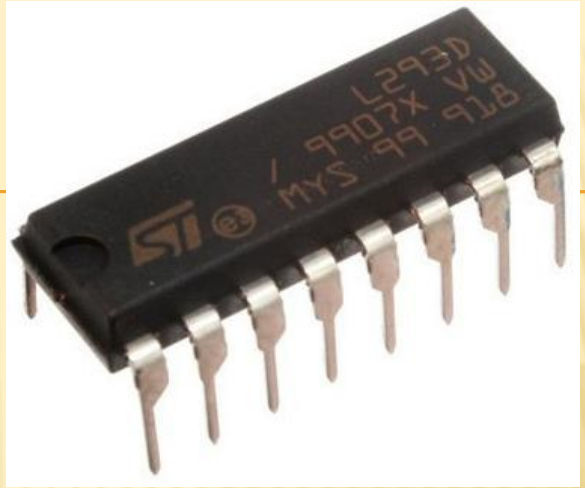
SW1,SW4 represents a half-H-Bridge
Half H-Bridge can output either Vcc or GND (0V)
To form a full H-bridge, we need two half H-Bridges



- ✓ IN=0V → SW4 is closed
- ✓ IN=5V → SW1 is closed



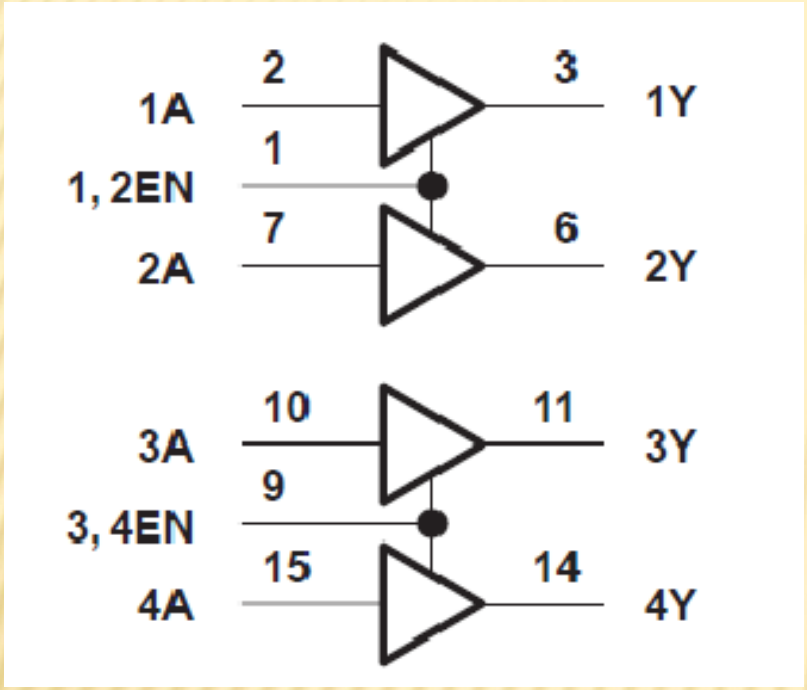
L293D TWO FULL H-BRIDGE

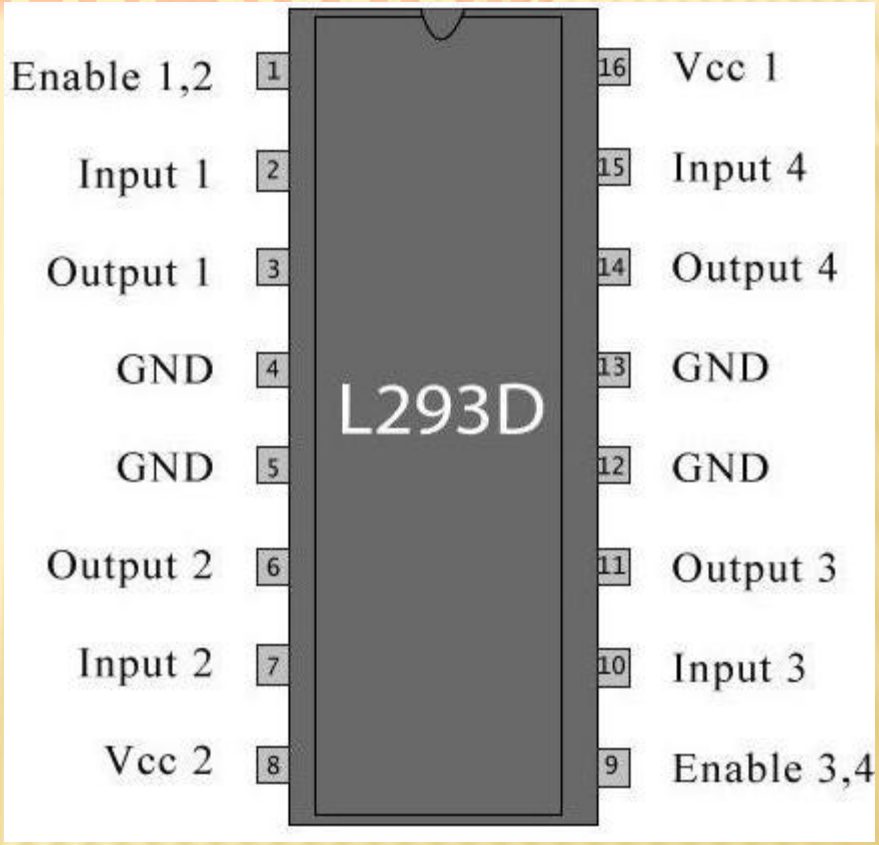


L293D FOUR HALF H-BRIDGE

- L293D is an integrated circuit for DC Motor Drive
- It is used with microcontroller to run dc motor forward and backward
- It can control two motors and their directions to turn clockwise or anti-clockwise
- it needs 3 singles from microcontroller to control one to enable it and two others for inputs

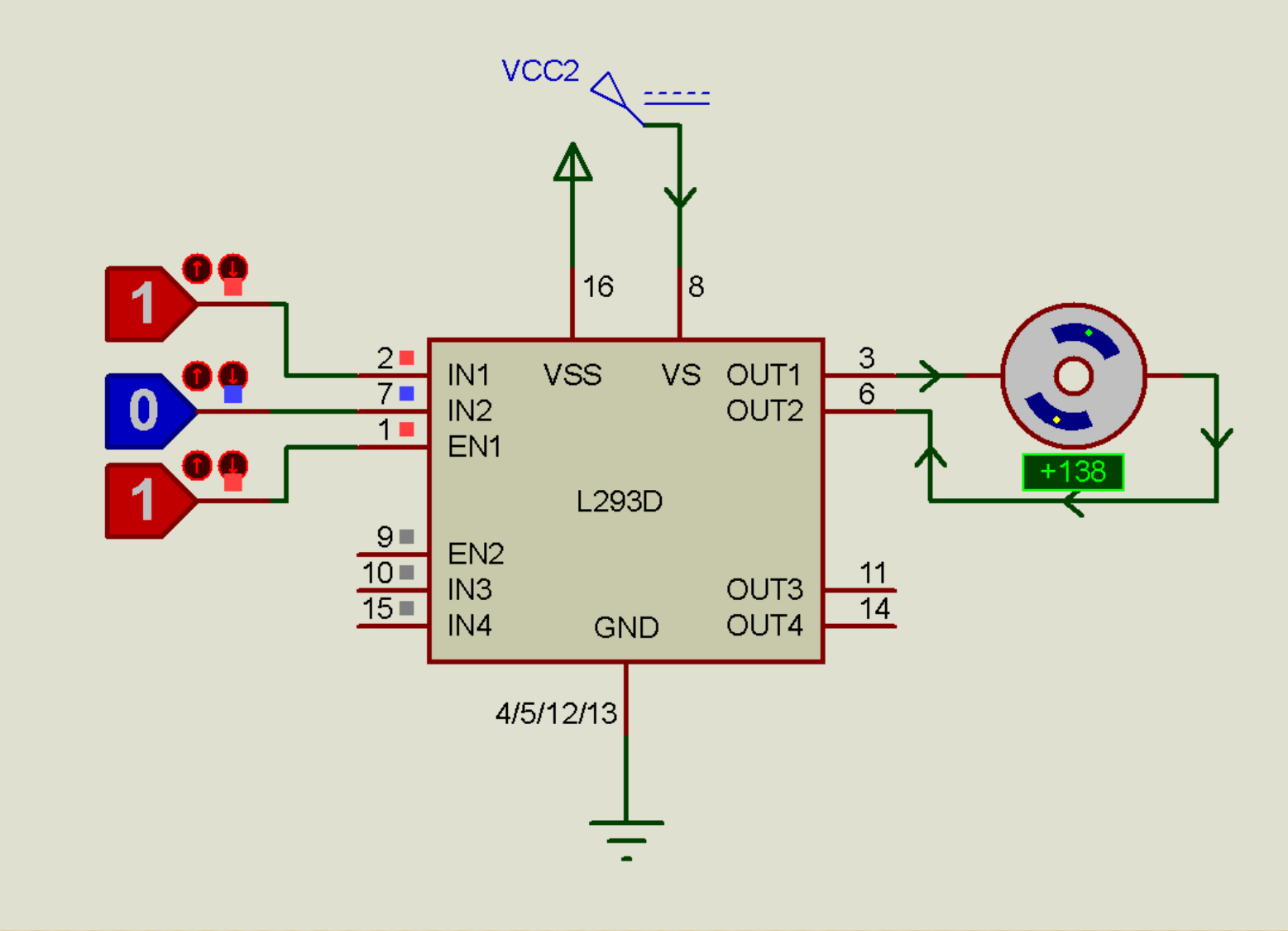
L293D BIN CONFIGURATION



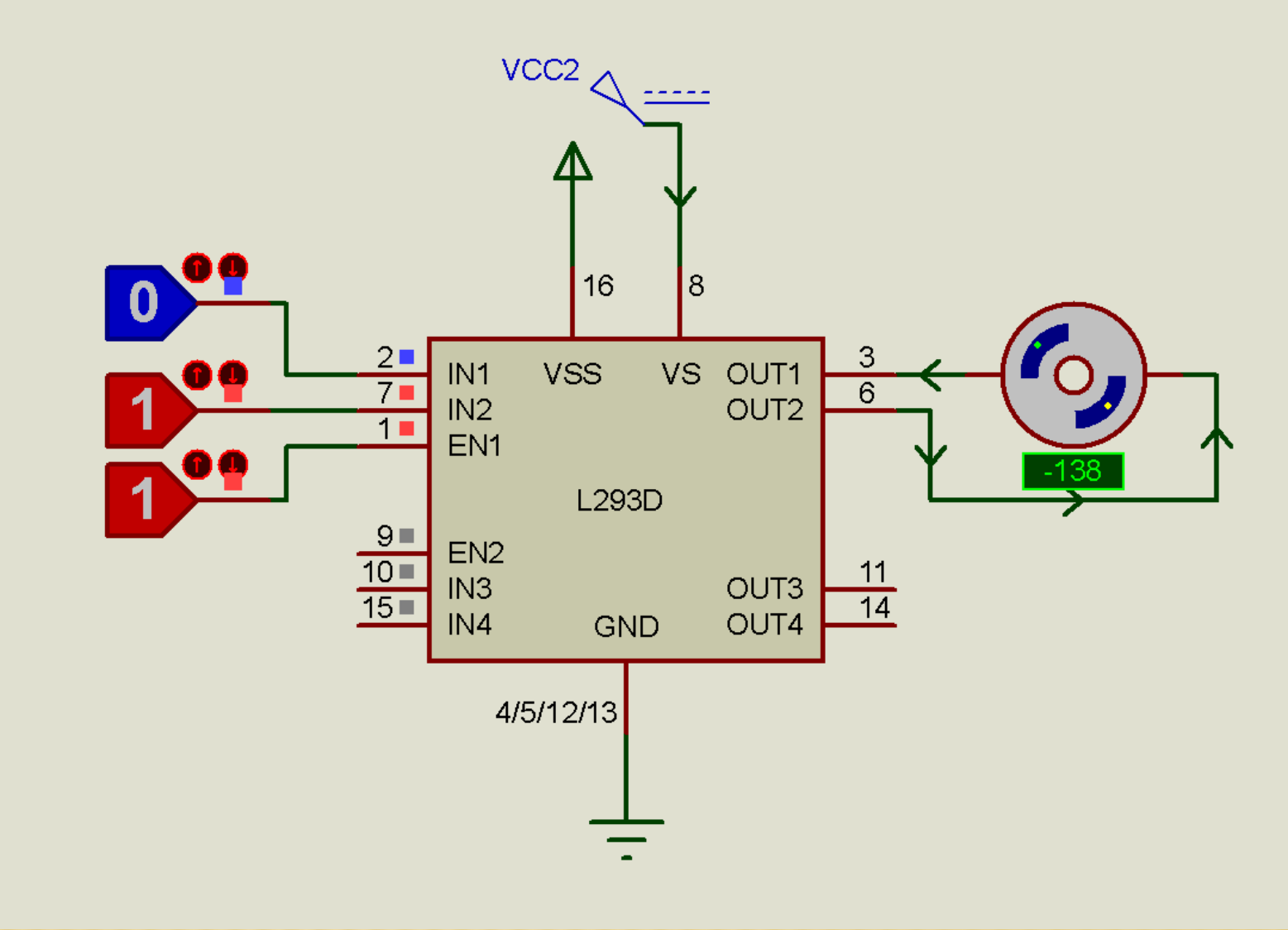


- ✓ VCC1→ Logic Supply voltage (5V)
- ✓ VCC2 → Motor Supply Voltage (12V)
- ✓ Enable 1,2 → Turn ON/OFF Half H-bridge 1,2
- ✓ Input1 → input of half H-bridge 1
- ✓ Output1→ output of half H-bridge1

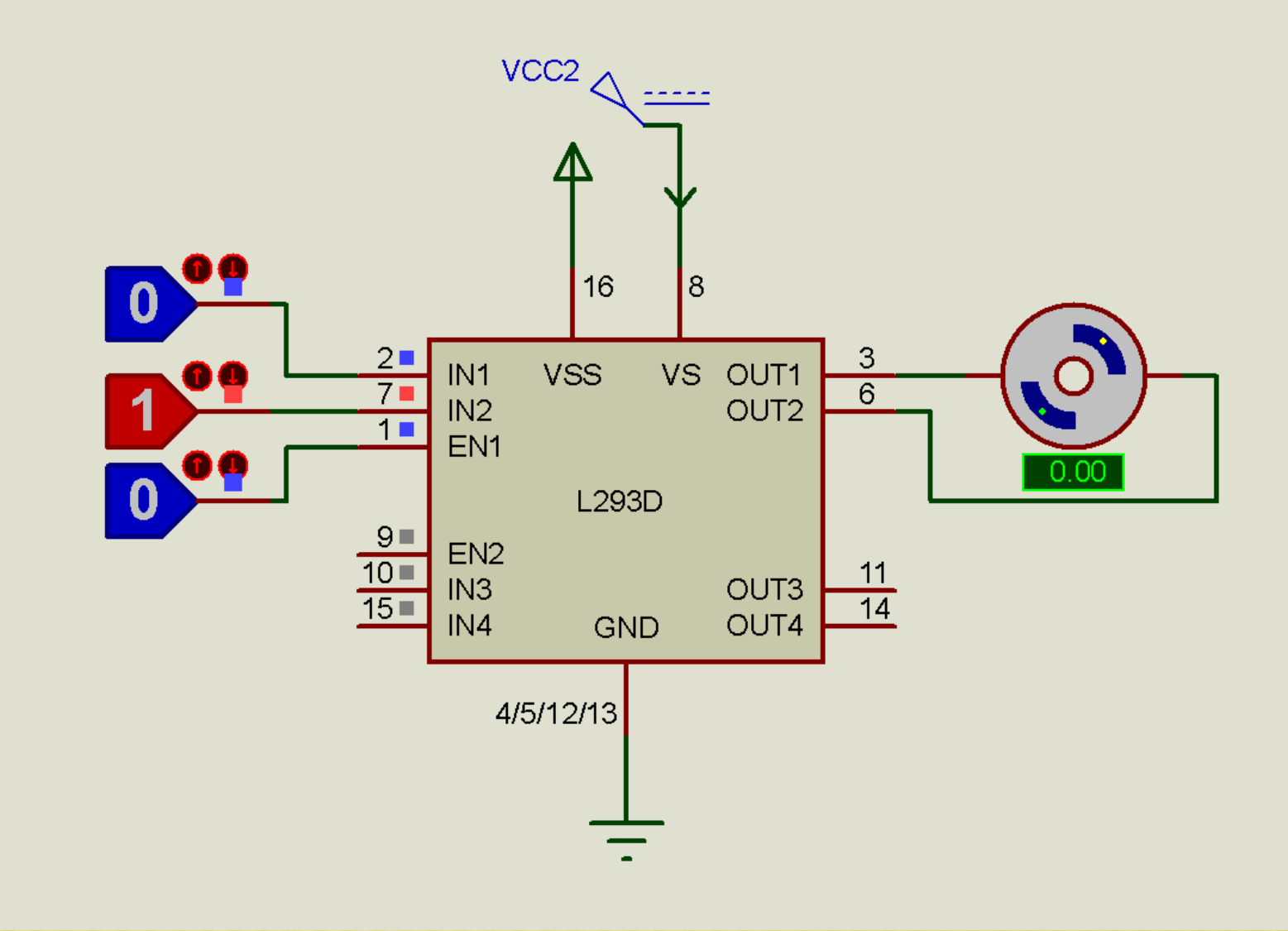
L293 FORWARD DRIVE



L293 BACKWARD DRIVE



L293 FORWARD DRIVE



DC MOTOR SPEED CONTROL

- In Robotics applications, the motor speed must be controlled
- In some tasks we need high speed and other tasks need low speed
- There are tasks where the motor speed varies along the time
- So How DC-Motor speed can be controlled
- There are to strategies based on:
  - ✓ By increasing or decreasing the effective voltage applied to the motor.
    - ➢ High voltage makes motor rotates faster;
    - ➢ Low voltage makes motor slows down.

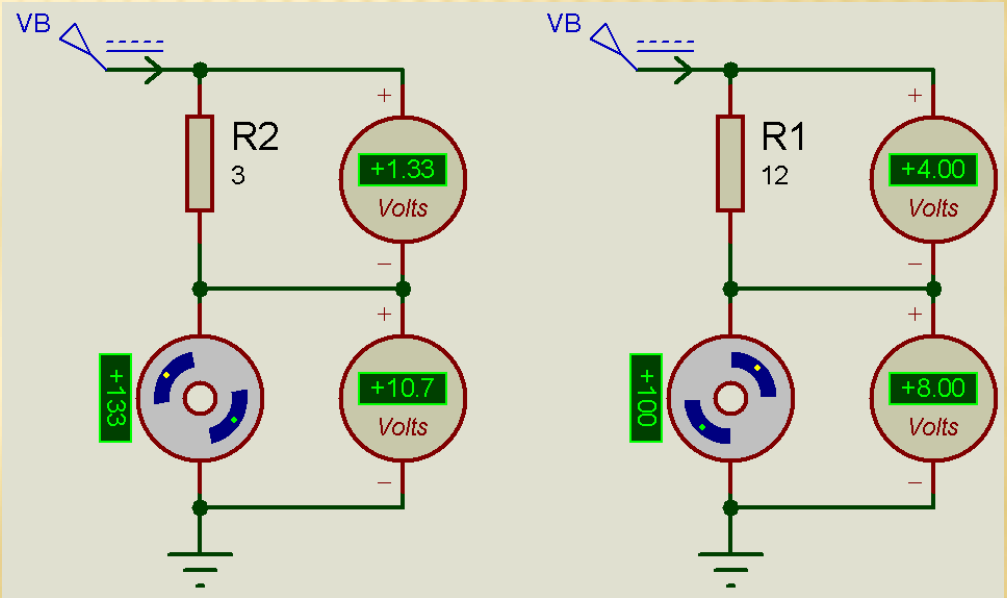


DC MOTOR SPEED CONTROL

- There are two ways to control the power delivered to a motor
  1) Using resistor as voltage divider
  2) Using Pulse Width Modulation (PWM)
- In both methods the effective voltage/power is controlled (reduced)

1) Method 1 Using Resistor

- A resistor is used to reduce the voltage/current delivered to the motor
- As resistance decreases more current/voltage delivered to motor and its speed increases
- This method is not practical where

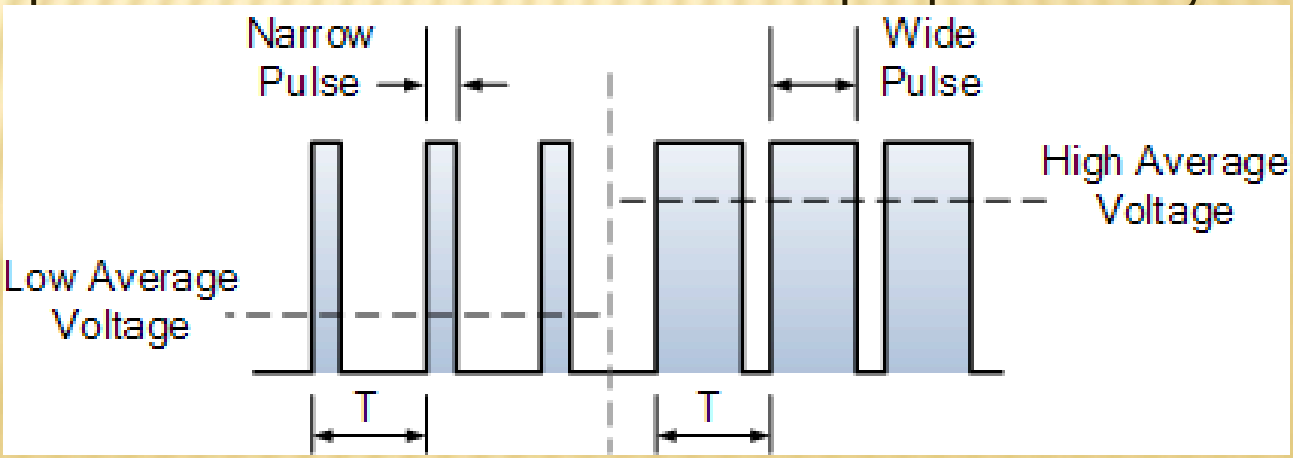


1. Resistor will consume large power and decrease efficiency
2. High power resistor is big and need large space

DC MOTOR SPEED CONTROL

2) Method 2 Using PWM

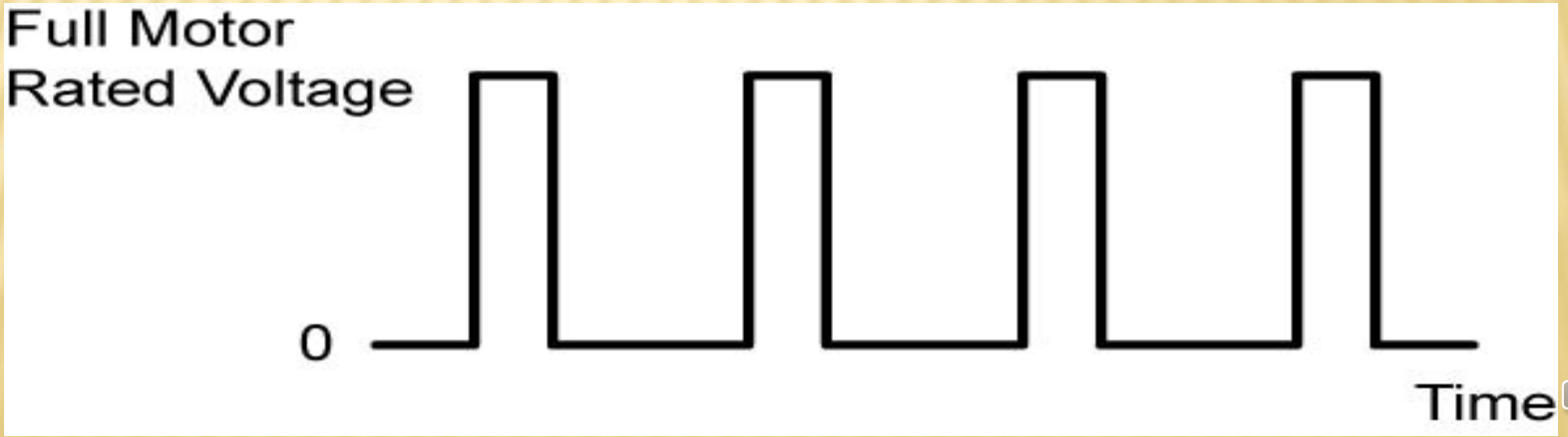
- It is an effective method for adjusting the amount of power delivered to an electrical load.
- It can be used to control speed, brightness, and power to many electrical device.
- It based on repeated POWER ON/POWER OFF sequence with high frequency and by
- Varying the ratio between POWER ON/POWER OFF times, the average power delivered is also varied proportionally



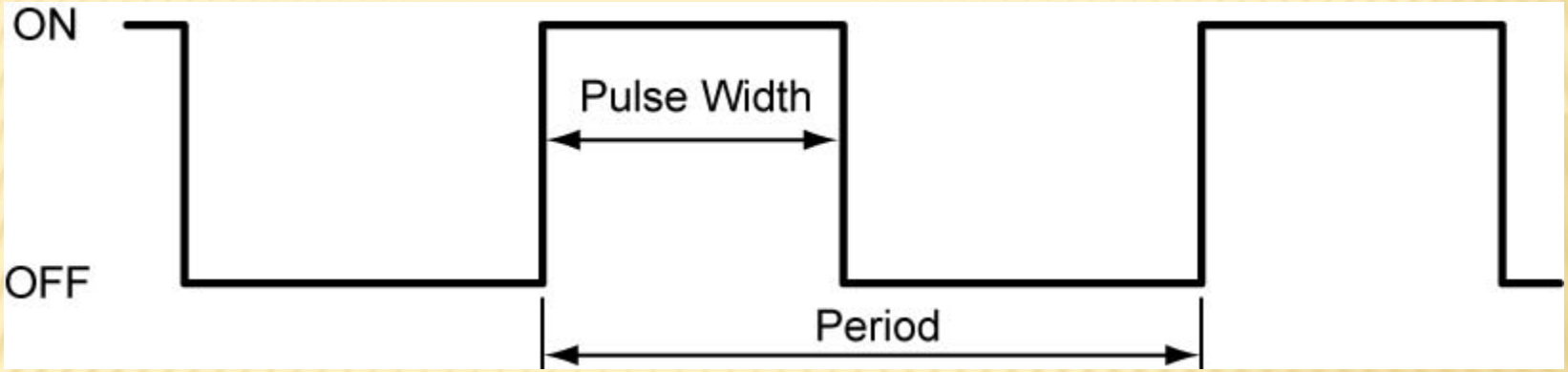
DC MOTOR SPEED CONTROL

2) Method 2 Using PWM

- PWM produces digital pulses at the full motor rated voltage
- By varying the pulse width in software, we also change the effective motor voltage and therefore control the motor speed.
- A microcontroller-based PWM solution uses fewer components and has the flexibility of varying the duty cycle/frequency on-the-fly through software

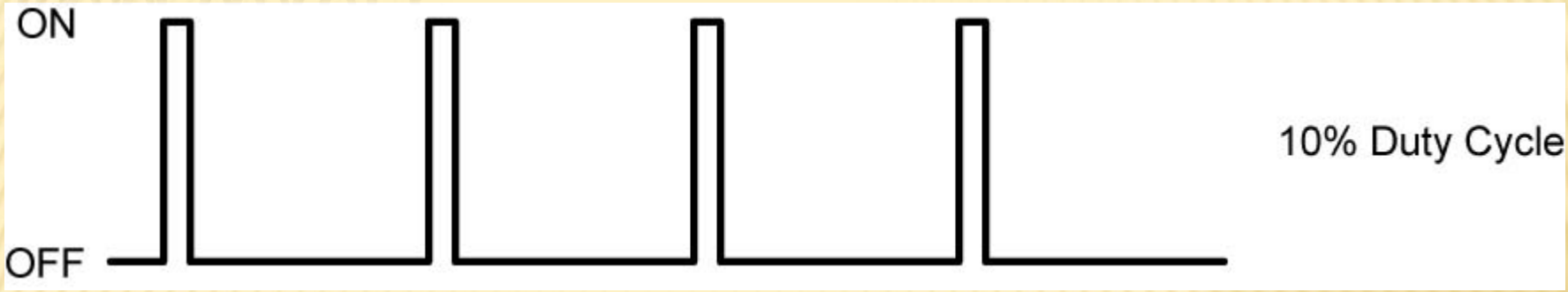


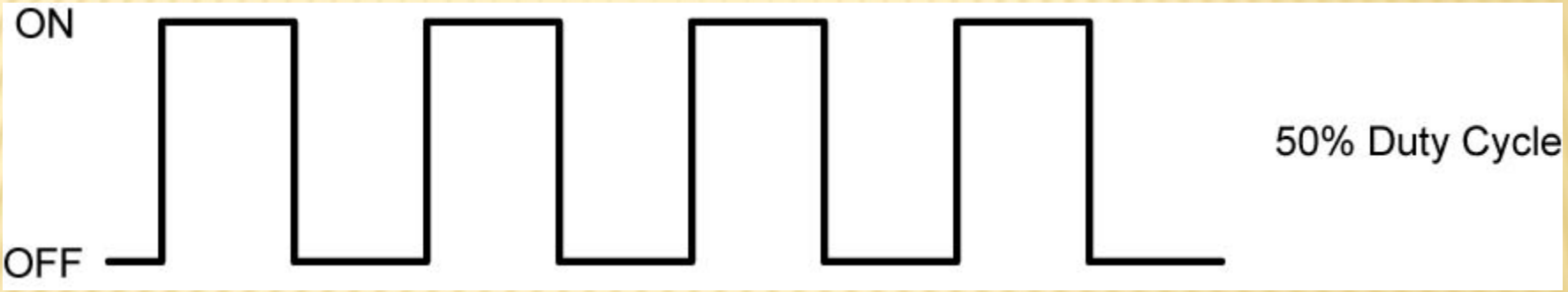
PWM PARAMETERS

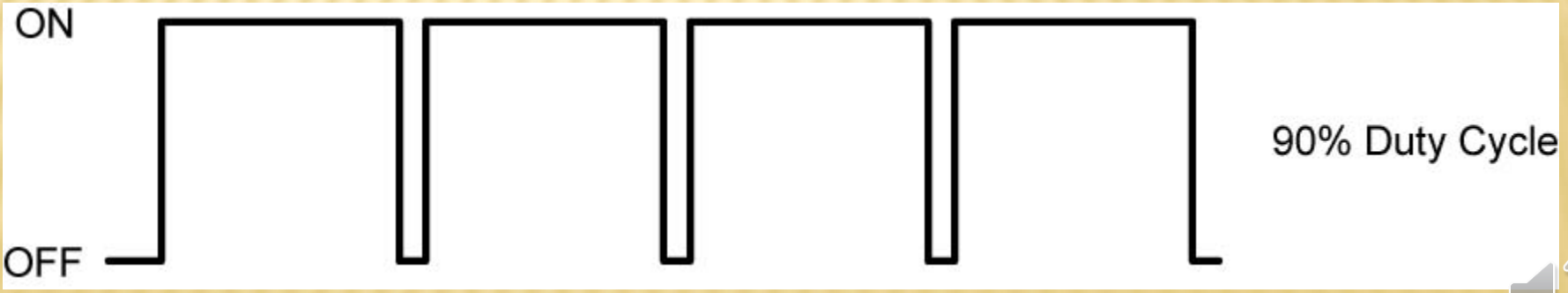


- The ratio $(T_{ON}/T_{period})\%$ is called the "pulse–width ratio" or duty cycle.
- When the duty cycle is 0%, the load (motor) is fully off;
- When the duty cycle is 100%, the load is fully on.

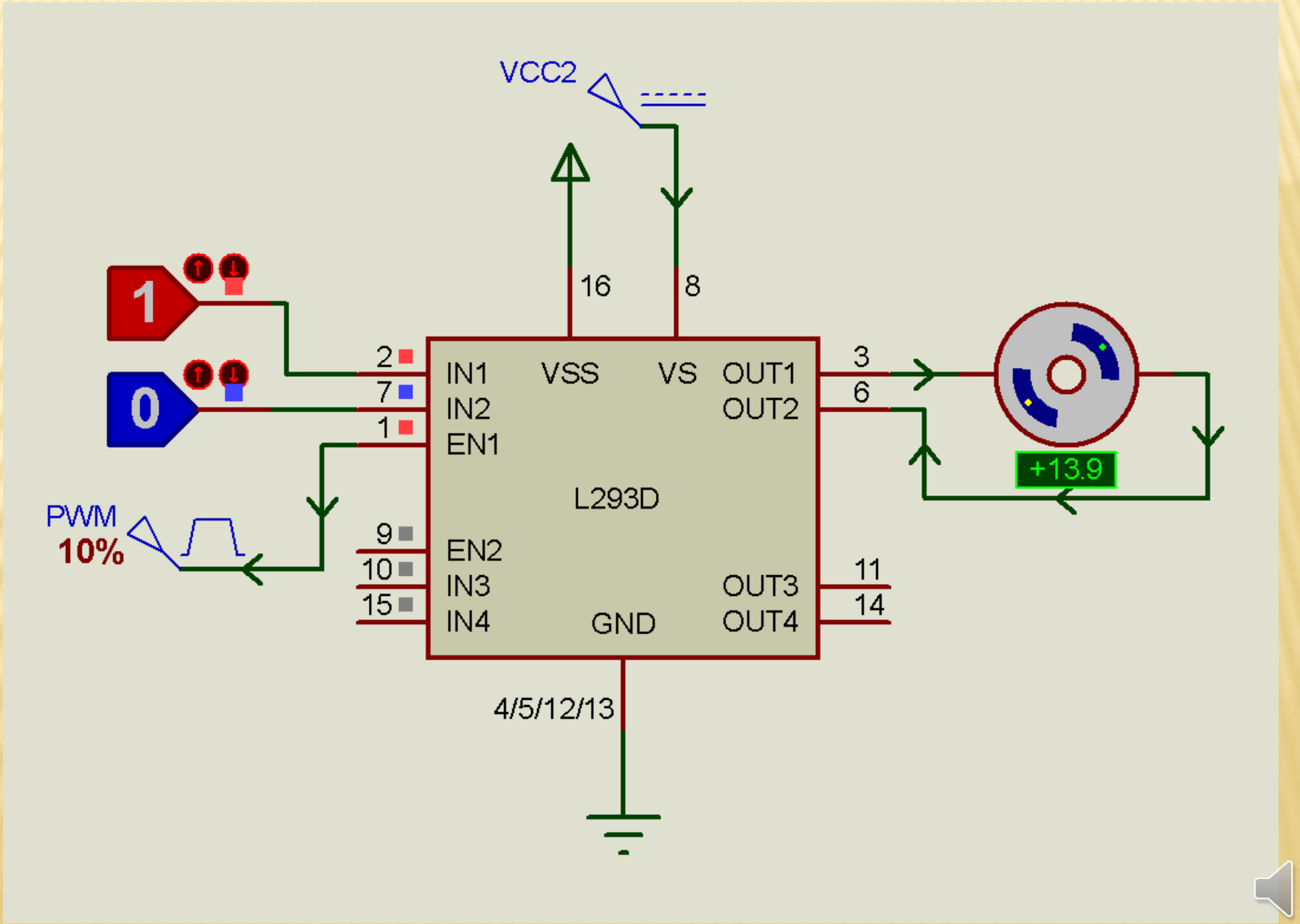
PWM WAVES



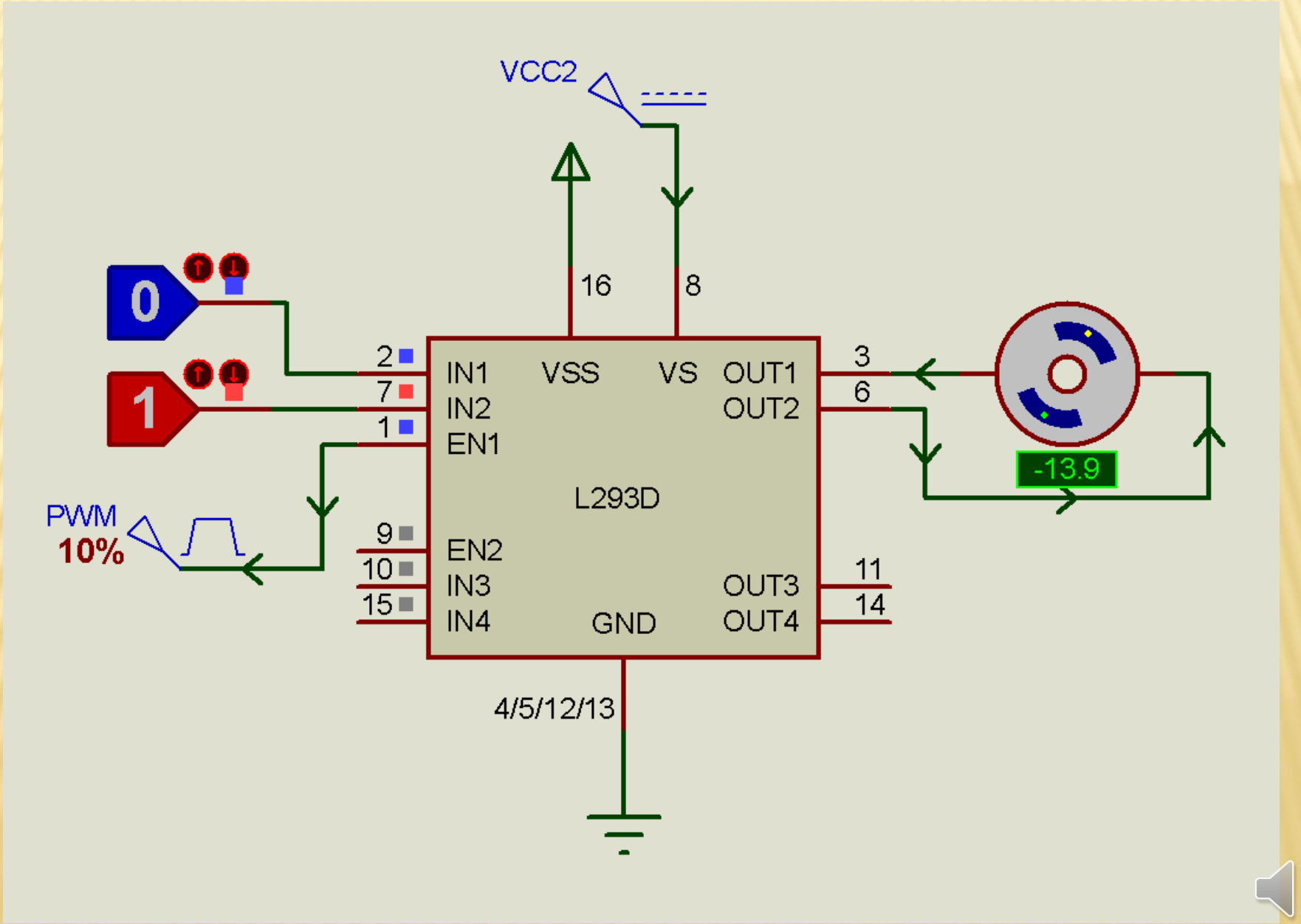




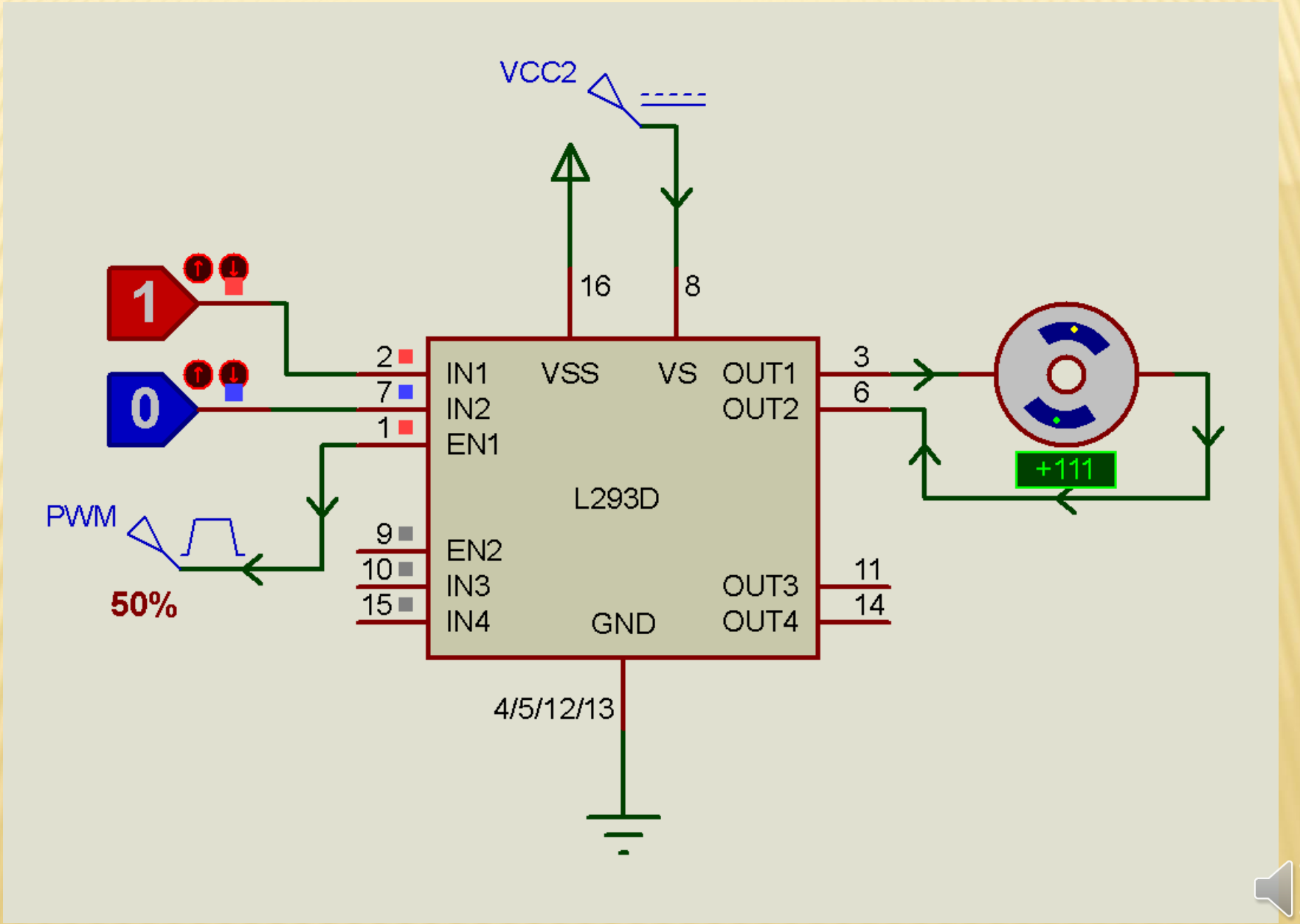
SIMULATION EXAMPLE1 FORWARD 10%



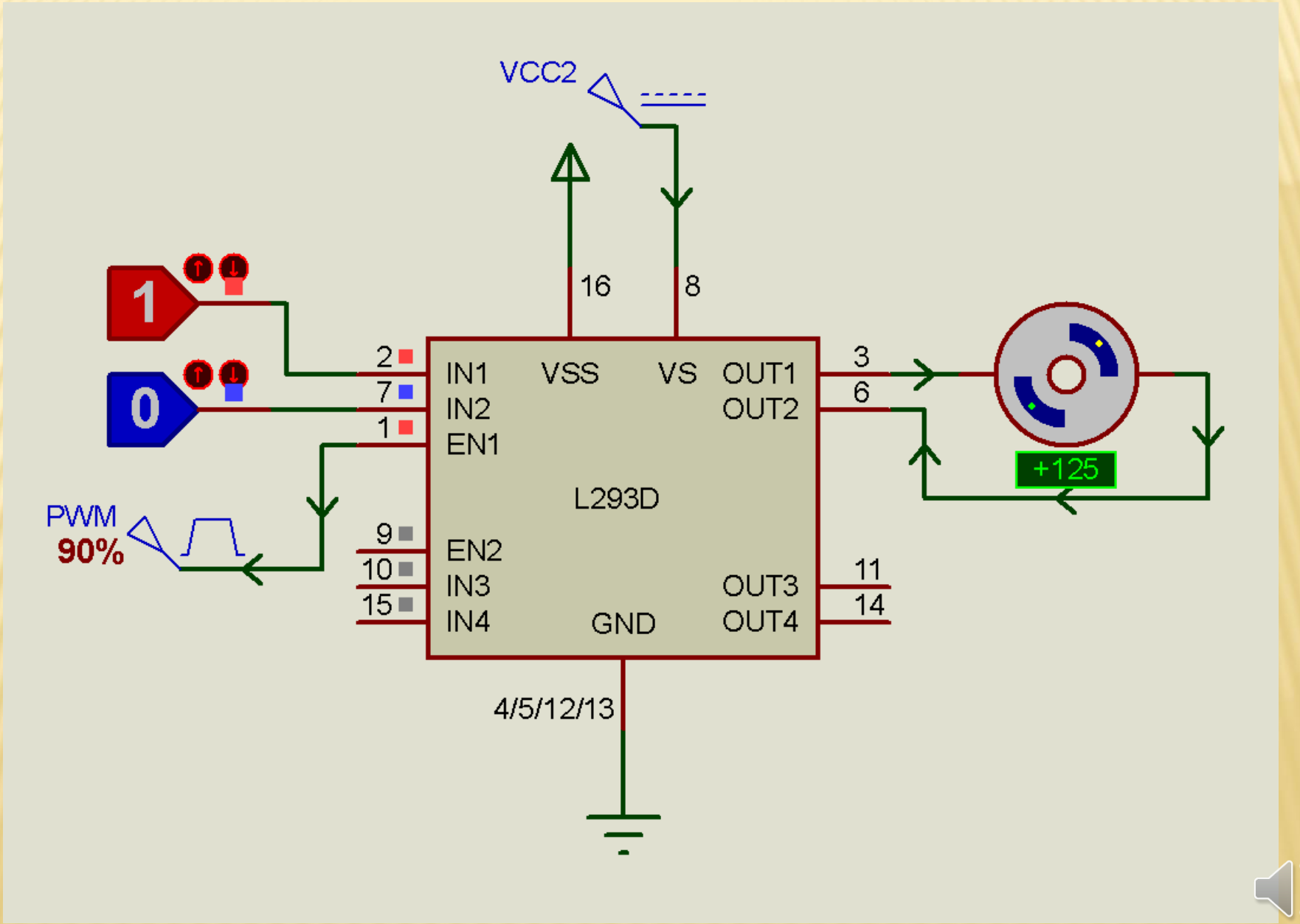
SIMULATION EXAMPLE2 BACKWARD 10%



SIMULATION EXAMPLE3 FORWARD 50%

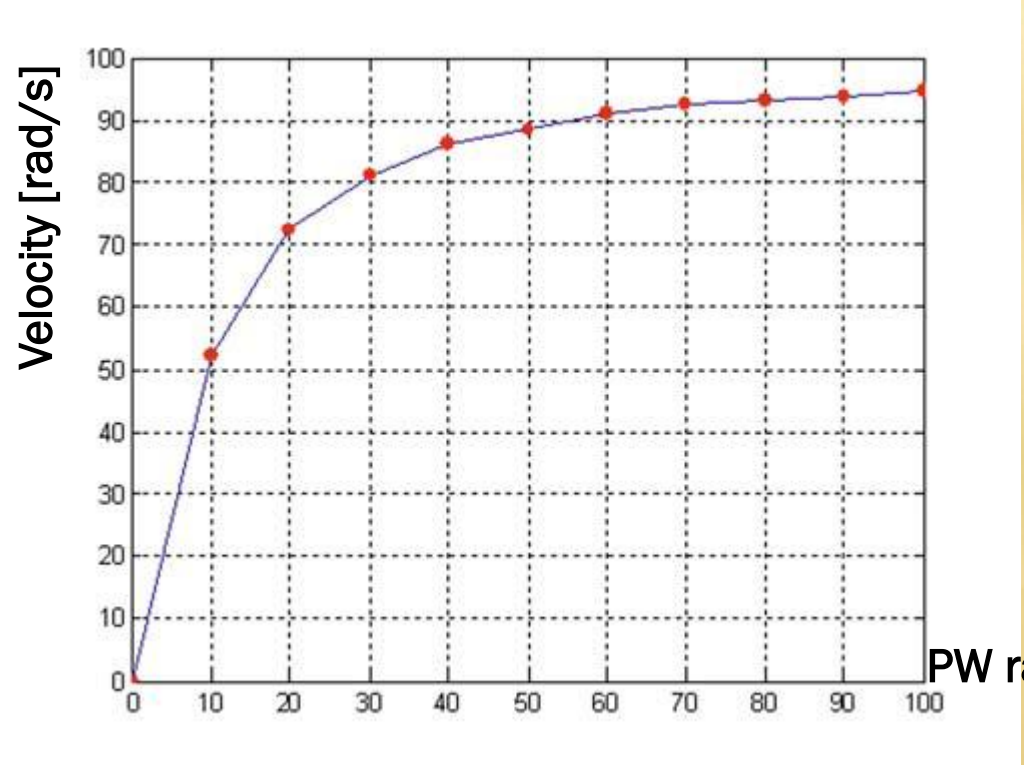


SIMULATION EXAMPLE4 FORWARD 90%



MOTOR CALIBRATION

- The relation between duty cycle and motor speed is not linear
- So we need to calibrate motors used in robots to define the relation between speed and duty cycle
- Motor calibration is done by measuring the motor speed at various settings of duty cycle between 0 and 100,
- This is usually recorded in in a motor calibration table



IS PWM DEFINE ACTUAL MOTOR SPEED

Finally , We are now able to achieve the two goals

1. Drive a motor forward or backward
2. Change its speed.

However, we have no way of telling at what speed the motor is actually running for a given duty cycle.

Note

The actual motor speed does depend not only on the **PWM** signal supplied, but also on **external factors** such as the load applied (for example the weight of a vehicle or the roughness of ground/floor)

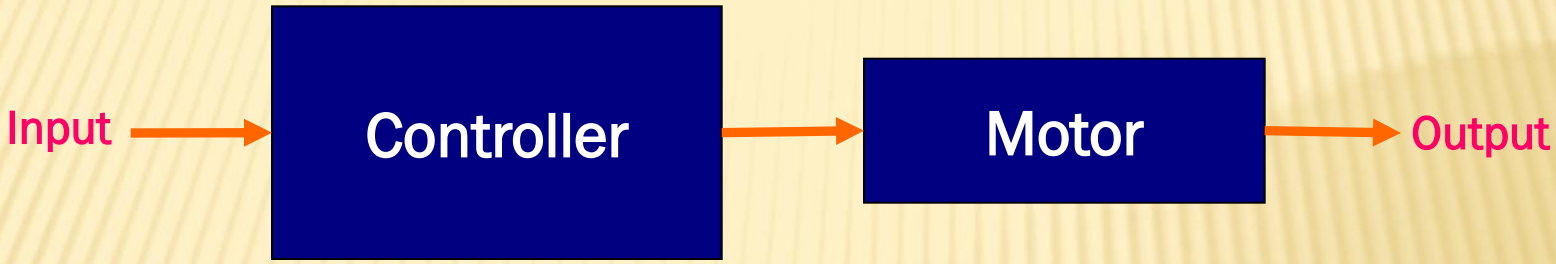


ACTUATOR CONTROL

- Robots are classified by control method into **servo** and **non-servo** robots
- Non-servo robots are essentially **open-loop** devices whose movements are limited to predetermined mechanical stops
- Servo robots use **closed-loop** control to determine their motion



OPEN LOOP CONTROLLER



- An **open-loop controller** (or non-feedback controller) is a type of controller which computes its input into a system using only the current state and its model of the system
- The system does not observe the output of the processes that it is controlling

OPEN LOOP CONTROLLER (CONT…)

- An open-loop controller is often used in simple processes because of its simplicity and low-cost, especially in systems where feedback is not critical
- Generally, to obtain a more accurate or more adaptive control, it is necessary to feed the output of the system back to the inputs of the controller

